

LAB 9

Wireless Network Setup and Packet Analysis using Cisco Packet Tracer

OBJECTIVES

- To configure a wireless router and establish a functional wireless network in Cisco Packet Tracer.
- To connect multiple wireless devices to the network and assign IP addresses dynamically using DHCP.
- To implement WPA2 security settings to ensure secure wireless communication.
- To analyse and verify wireless packet transmission using Packet Tracer's simulation mode.

THEORY

1. Wireless Networking Fundamentals

Wireless networks allow devices to communicate without physical cabling, using radio frequency (RF) signals. A wireless router acts as the central device, providing connectivity to laptops, smartphones, and other wireless clients. The IEEE 802.11 standards define the protocols for Wi-Fi communication, ensuring interoperability across devices.

2. Role of the Wireless Router

The wireless router integrates three essential functions:

- **Access Point (AP):** Provides wireless connectivity to client devices.
- **Router:** Directs traffic between the local network and external networks.
- **DHCP Server:** Automatically assigns IP addresses to connected devices, simplifying configuration.

In Cisco Packet Tracer, the WRT300N router model is commonly used to simulate wireless networking scenarios.

3. Wireless Security – WPA2

Wireless networks are vulnerable to unauthorized access if left unsecured. WPA2 (Wi-Fi Protected Access 2) is the industry-standard security protocol that uses AES (Advanced Encryption Standard) for strong encryption. Configuring WPA2 ensures:

- Authentication of legitimate users.
- Confidentiality of transmitted data.
- Protection against common attacks such as eavesdropping and unauthorized access.

5. Packet Analysis in Cisco Packet Tracer

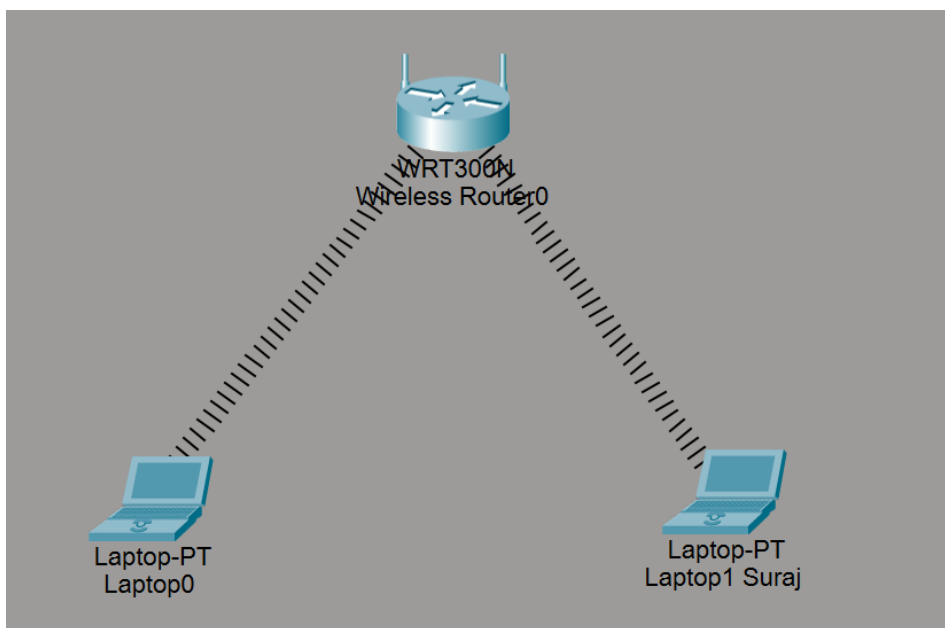
Packet Tracer's simulation mode allows visualization of data flow across the network. By analysing packets, students can observe:

- DHCP request and acknowledgment exchanges.
- ARP (Address Resolution Protocol) operations.
- Encrypted wireless communication under WPA2.

This reinforces theoretical concepts with practical, hands-on verification

NETWORK TOPOLOGY

The topology used in this lab is a basic wireless star topology, where the wireless router acts as the central hub and all devices connect to it. Each laptop communicates with the router through wireless signals, ensuring centralized management of IP addressing and security. This setup reflects real-world home or small office networks, where multiple clients rely on a single wireless access point for connectivity.



CONFIGURATION

THE DHCP configured IP of Laptops are as follow:

Device	IPv4 address	Subnet Mask	Default Gateway
Laptop0	192.168.0.101	255.255.255.0	192.168.0.1
Laptop1 Suraj	192.168.0.102	255.255.255.0	192.168.0.1

Procedure :

Wireless Router Configuration :

Step 1 : Basic Setup

1. Click on the Wireless Router.
2. Go to GUI → Setup.
3. Configure the following:
 - IP Address: 192.168.0.1
 - Subnet Mask: 255.255.255.0
 - DHCP Server: Enabled

Step 2 : Wireless Configuration

1. Go to Wireless → Basic Wireless Settings.
2. Configure the following parameters:
3. Network Mode: Mixed
 - SSID: HCOE
 - Channel: Auto
4. Click Save Settings.

Step 3 : Wireless Security

1. Go to Wireless → Wireless Security.
2. Configure the following:
 - Security Mode: WPA2-Personal
 - Encryption: AES • Passphrase: hcoe@123
3. Save the configuration.

TESTING

1. Testing if both laptops connected to WirelessRouter0



Fig: Laptop successfully connected to access point (Wireless Router)

2. Ping Test (From Laptop1 to Laptop0)

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Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.101

Pinging 192.168.0.101 with 32 bytes of data:

Reply from 192.168.0.101: bytes=32 time=32ms TTL=128
Reply from 192.168.0.101: bytes=32 time=13ms TTL=128
Reply from 192.168.0.101: bytes=32 time=17ms TTL=128
Reply from 192.168.0.101: bytes=32 time=15ms TTL=128

Ping statistics for 192.168.0.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 13ms, Maximum = 32ms, Average = 19ms
```

Fig: Ping test

DISCUSSION

The wireless network was successfully configured in Cisco Packet Tracer using the WRT300N router and laptops as client devices. DHCP ensured automatic IP assignment, reducing manual configuration errors. WPA2 security settings provided encrypted communication, demonstrating the importance of securing wireless networks against unauthorized access. Packet analysis in simulation mode confirmed proper DHCP exchanges and secure data transmission, validating the theoretical concepts applied in practice.

CONCLUSION

This lab demonstrated the setup of a wireless network in Cisco Packet Tracer, highlighting DHCP's role in dynamic IP allocation and WPA2's importance in securing communication. The packet analysis verified correct operation of the network, reinforcing both theoretical understanding and practical skills in wireless networking.